Católica School of Business and Economics

Business Analytics - Foundation of Statistics with Applications in R

Group Submission

Project Scoping Package

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Data Familiarization

All seven tables were explored in R (cross-checked in SQL). Key findings per table are summarised below; the table relationships are shown in Appendix 1.

Customers: 82,570 rows. The primary key (PK) customer_record_id identifies a customer for a specific order (1 : 1 with orders). customer_id identifies the person and is not unique (79,759 people), so only ~3% of customers are repeat buyers. Location is stored per record and is highly concentrated: 42% of records are in one region.

Orders: 82,570 rows. The PK order_id links to order_items, payments and customer_feedback, and customer_record_id links to customers. Missing timestamps are mostly structural (orders that never shipped), but the logic is not always respected: 5 cancelled orders have a delivery date, 7 completed orders have none, and 1,127 orders break the sequence purchase < approval < carrier < delivery. We also found 21 groups of possible duplicate orders (same customer, second and product). Timestamp-based variables therefore need careful cleaning.

Products: 29,389 rows. The PK product_id links to order_items, with no duplicates. Category, name/description length and photo count are missing together for 533 products (incomplete registration); category labels are inconsistently spelled (78 labels for 73 categories) and 10 weights are implausible (0 g or above 100 kg).

Payments: 86,345 rows. An order can be paid in several parts, so the key is order_id + payment_sequence: 2,478 orders have several payments and 71 have gaps in the sequence. Payments must be summed per order before joining; they then match item price plus freight for 99.7% of orders.

Customer_feedback: 82,380 rows. Only the pair feedback_id + order_id is unique: 646 feedback_ids cover several orders and 166 orders have conflicting ratings, so one rating per order must be chosen. Ratings are J-shaped (58% five stars, 11.5% one star).

Sellers: 2,983 rows. The PK seller_id links to order_items; there are no missing values or duplicates and every seller has sales. Sellers are even more concentrated than customers (59% in one region).

Order Items: 93,519 rows, one per item line. It is the bridge table linking orders (order_id), products (product_id) and sellers (seller_id). order_id + item_number should be the PK, but

60 rows are exact duplicates; payments only match once they are removed, confirming they are errors. 1,052 orders involve more than one seller.

An overview of how the tables are linked is provided in the appendix

Proposed Project Scope

Research Question: *Are orders where seller and customer are in the same region delivered faster and rated higher, and how much of the rating difference is explained by shorter delivery times?*

Business Perspective Relevance: Delivery is the part of the purchase Aurora Market can influence most directly, and ratings shape trust in the platform. Knowing how much proximity and delivery time affect satisfaction shows whether optimising the shipment process (e.g. strengthening the seller base in regions mostly served from far away) is a credible lever for satisfaction and, ultimately, sales.

Unit of observation: The order. The sample consists of completed, delivered orders with a rating (≈80,000). Customers are joined through customer_record_id, and item-, payment- and feedback-level tables are aggregated to one row per order before joining.

Likely Outcome & Explanatory Variables: We expect same-region shipments to be faster and better rated, with the rating gap largely explained by shorter delivery times.

Explanatory Variables	
Variable	Desc
Delivery Time (DelT)	Amount of time the shipping process took.
Freight (Freight Cost)	Total shipping costs.
SameRegion	This would be a dummy, Represents if the sellers and customer belong to the same region.
EstimatedDiff	Difference between the actual shipping date and the estimated shipping date
NoSellers	Amount of sellers the order was related to.

Tables we expect to use: orders (spine), order_items, sellers, customers, customer_feedback, products and payments. Joins: customer_record_id (orders–customers, 1 : 1); order_id (orders–order_items, customer_feedback, payments); seller_id and product_id (order_items–sellers, products). Tables with several rows per order are aggregated first, since naive joins duplicate rows.

Analytical/Statistical approach: We estimate the equation using multiple linear regression (OLS) at the order level, explaining customer rating (1–5) by delivery time, freight cost, a same-region dummy, the gap between actual and promised delivery date, and the number of sellers. Each coefficient shows the change in rating from a one-unit change in that variable, holding the others constant, and including both delivery time and the promise gap separates slow delivery from late delivery. We interpret coefficients by sign, size, confidence interval, and p-value. It's also worth noting that our dummy will provide us with 2 curves representing the different behavior within same /different region customer/sellers.

$$Rating = \beta_0 + \beta_1 DelT + \beta_2 Freight + \beta_3 SameRegion + \beta_4 EstimatedDiff + \beta_5 NoSellers + u$$

Limitations/Data issues/Analytical risk:

Missing geographical detail: locations are anonymized codes, with no distances or infrastructure (roads, airports, logistics hubs). These affect delivery time and are likely correlated with same_region, causing omitted-variable bias: same_region may partly capture better infrastructure rather than proximity. Region dummies absorb differences between regions, not within them.

Limited customer characteristics: no demographics (age, income) and almost no purchase history (97% buy once). If rating harshness differs by location, estimates are biased; ratings are also only observed for customers who leave feedback.

A few regions dominate: one region holds 42% of customers, 59% of sellers and 88% of same-region orders, and within it the rating gap is small (4.25 vs 4.23); small regions give imprecise estimates. We will check results hold with and without the dominant region.

Appendix 1:

